Nabu: A Real-Time Vocal AI Coach

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***Abstract*—Mental well-being has become a critical global concern, yet access to professional therapy is often limited by high costs, long waiting times, and persistent social stigma. Many individuals hesitate to share emotional distress, highlighting the need for private, accessible, and non-judgmental support systems. In this context, AI-based coaching offers a promising alternative by providing immediate and discreet assistance. This paper presents Nabu, a hardware-integrated AI coaching system designed to provide personalized emotional support and behavioral guidance. Unlike software-only solutions, Nabu incorporates a physical embodiment to enhance user engagement and trust. The system combines natural language processing (NLP) with a therapy-trained large language model (LLM) to act as a conversational companion, offering real-time coaching grounded in psychological frameworks. A key innovation of Nabu is its ability to analyze vocal patterns and emotional cues—such as pitch and tone—without requiring users to explicitly state their feelings. Based on affective computing principles, this enables proactive and adaptive support. The hybrid architecture integrates the local processing power of the Hailo-8 accelerator with advanced linguistic capabilities, ensuring low-latency, privacy-preserving, and context-aware interactions. To encourage user comfort, the "Nabu's Room" interface incorporates human-like signals that reduce perceived technological distance. The modular design also aligns with the principles of social robotics, emphasizing accessibility and emotional connection. Overall, Nabu demonstrates the potential of hybrid AI systems to function as empathetic companions, offering secure, responsive, and emotionally intelligent support for users.**

***Keywords*—Vocal Analysis, AI Coaching, Real-Time Systems, Affective Computing, Natural Language Processing, Human-Computer Interaction.**

I. Introduction

AI-based coaching provides a non-judgmental alternative for individuals who hesitate to share emotional distress within their social circles [1].

The Nabu system addresses this need by utilizing the principles of *Affective Computing* established in [2]. By analyzing vocal patterns such as pitch and tone, the system detects emotional nuances without requiring explicit disclosure, enabling a proactive and empathetic response.

The remainder of this paper is organized as follows: Section II reviews *Related Work*; Section III describes the *System Design* and hardware; Section IV details *Implementation and Testing*; Section V summarizes the *Conclusions*.

II. Related Work

To evaluate the position of Nabu within the current technological landscape, this section analyzes the State of the Art (SotA) in emotional AI and the existing Gold Standard systems for mental health assistance.

A. State of the Art (SotA) in Affective Computing

The current State of the Art in vocal emotion recognition focuses on deep learning architectures capable of extracting prosodic and spectral features in controlled environments. As established in [3], intelligent audio analysis has reached a point where nuance detection (e.g. subtle stress or fatigue) is possible through high-dimensional feature vectors. However, most SotA models are computationally expensive and typically operate in an "offline" mode, meaning the analysis happens after the recording ends. The gap in current SotA lies in the transition from laboratory accuracy to real-time, embedded low-latency applications that can provide immediate feedback.

B. The Gold Standard in Digital Mental Health

In the clinical and consumer market, the "Gold Standard" for digital interventions is currently represented by text-based Cognitive Behavioral Therapy (CBT) agents, such as Woebot [4] or Wysa [5]. These systems are considered the benchmark because they follow strict psychological protocols and have been validated through clinical trials.

However, these Gold Standard systems have significant limitations:

- *Input Barrier:* They rely on manual text input, which can be exhausting for users in acute distress.
- *Loss of Context:* Text cannot capture the physiological indicators of emotion (trembling voice, pitch shifts) that are identified in [6] as crucial to building a persuasive and empathetic bond.

C. Bridging the Gap: The Nabu Approach

Nabu transcends the current Gold Standard by moving from text-based interaction to a real-time vocal loop. While existing research focuses on the *accuracy* of emotion detection, Nabu focuses on the *utility* of that detection within a coaching framework. By integrating the theoretical frameworks established in

[2] with a *lightweight* real-time vocal processing pipeline, the proposed solution offers a proactive alternative that detects emotional shifts before the user explicitly reports them.

III. System Design

The Nabu system is designed as a hybrid Edge-Cloud solution, balancing local processing for privacy and low latency with server-side computational power for complex emotional reasoning.

A. Overall System Architecture

The proposed architecture is based on a distributed Client-Server architecture, as shown in Fig. 1, which orchestrates the data flow between the user's physical environment and the analytical backend.

The operational process is divided into two main phases:

- *Session Initialization:* The process begins with a *User Login* and a *Fetch User Data* operation to personalize the experience. Once the user is authenticated, the *Session Initiation* command prepares the system for real-time interaction.
- *Data Exchange Loop:* During an active session, the *Client-Side* remains in a state of *Waiting for audio input*. Upon detecting voice, it transmits *Data packets* to the *Server-Side*. The backend simultaneously performs *Voice Analysis* and *Log Analysis* to generate a summarized response, ensuring that the coaching remains contextually consistent with previous interactions.

B. Advanced Sensor Fusion and Interaction Logic

A core innovation of the Nabu architecture is the implementation of *multi-modal sensor fusion* to achieve a granular understanding of the user's affective state. As illustrated in the operational logic pipeline (Fig. 2), the system transcends traditional audio-only analysis by correlating vocal prosody with physiological data from the heartrate sensor and kinetic patterns from the gyroscope.

This *sensor fusion* approach allows Nabu to distinguish between transient vocal fatigue and acute emotional distress, generating a "Stress Score" with higher fidelity than single-modality systems. By integrating these diverse data streams, the logic pipeline ensures that the Therapy-Trained LLM receives a holistic emotional context, facilitating highly personalized and adaptive feedback.

C. Hardware Architecture and System Integration

The physical implementation of the Nabu architecture is designed as an autonomous coaching station, as illustrated in Fig. 3. This setup is built around a *Raspberry Pi* and a *Hailo-8 AI accelerator*, enabling local processing for privacy-sensitive biometric data.

As established in [7], this edge-based approach is crucial for protecting user privacy by processing emotional cues at the source. The hardware (Fig. 3) further includes an *8-mic array* for noise cancellation, a *heartrate sensor*, and a *gyroscope* for

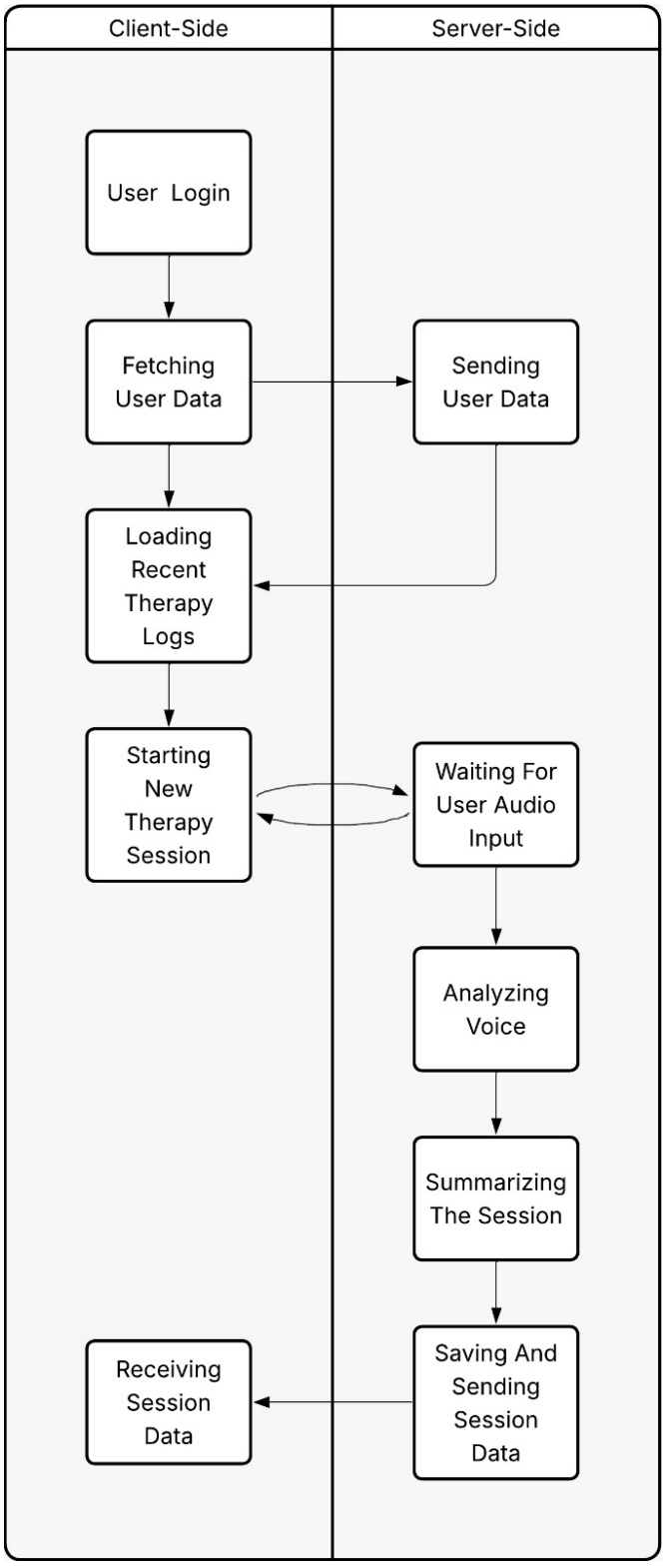


Fig. 1. High-level Client-Server architecture of the Nabu ecosystem.

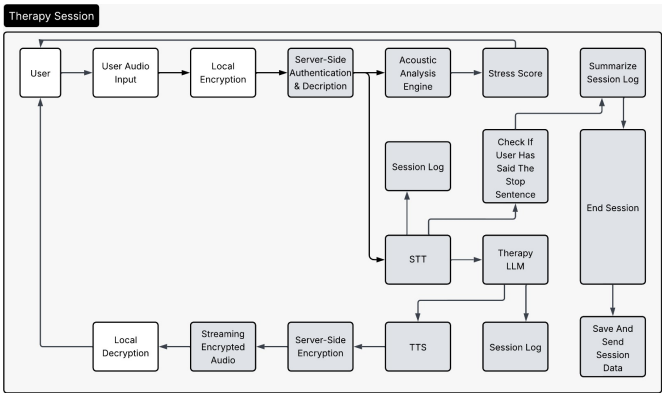


Fig. 2. Therapy session logical flow and emotional analysis pipeline.

multi-modal sensing. Stable operation is ensured by a custom PCB with a *Buck Converter* and active cooling.

While sensing is local, the *Server-Side* (Fig. 4) provides the high-level intelligence. It manages a *Therapy-Trained LLM* and a *Summarization Engine* that tracks user progress across sessions. To ensure data integrity, the backend follows the information security protocols defined in [8], creating a secure and persistent memory for the coaching process.

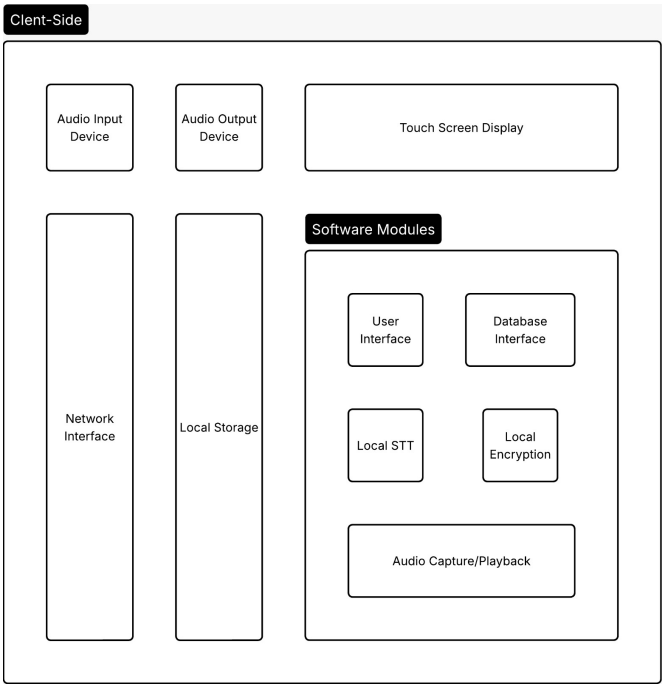


Fig. 3. Integration of physical hardware components and software layers.

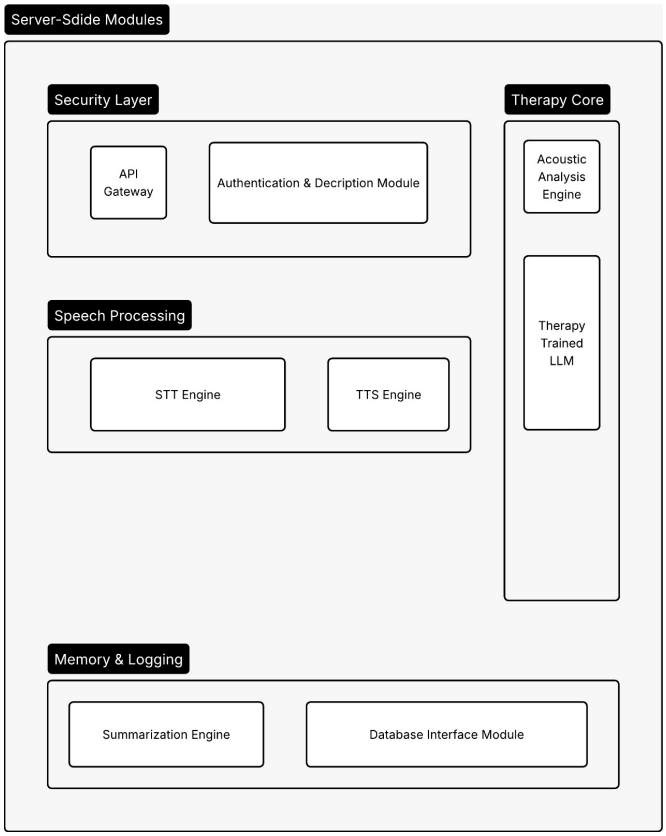


Fig. 4. Server-side functional modules and security layers.

IV. Implementation and Testing

A. Physical Prototype and Hardware Assembly

The physical realization of Nabu focuses on transforming a complex AI system into a friendly, approachable companion. The chassis was developed using *high-precision 3D printing* with a focus on minimalist design and ergonomic lines, ensuring it fits naturally into a modern living environment or a mindfulness-oriented setting, as seen in Fig.5.



Fig. 5. The Nabu physical prototype: A minimalist and friendly design intended for seamless integration into the user's personal space.

The assembly process prioritized structural modularity to ensure that each component remains accessible for maintenance. Beyond the external shell, the integration required

a strategic placement of the 8-microphone array to isolate it from the micro-vibrations of the internal cooling system. By implementing a custom-engineered acoustic buffer within the printed chassis, the system achieves high-fidelity sound capture while maintaining thermal equilibrium. This physical optimization ensures that the device remains silent and unobtrusive during sensitive conversations, preserving the intimate atmosphere of the interaction.

B. Software Stack and User Experience Design

The software layer is developed entirely in *Python*, chosen for its robust library support in AI and signal processing. The User Interface (UI), illustrated in Fig. 6, was designed to mimic the comforting atmosphere of a professional therapy session.

A key feature of the UI is the "Nabu's Room" aesthetic. Instead of a cold, technical dashboard, the user interacts with a friendly character in a visually warm environment. This "light" and "friendly" approach is intentional; it transforms the device from a piece of hardware into a supportive companion that "listens" and "speaks" like a friend. This design choice significantly lowers the user's anxiety, making the emotional interaction feel natural and safe.

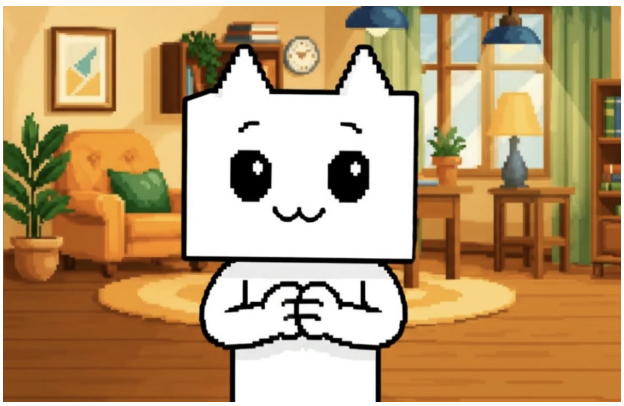


Fig. 6. The Nabu UI: A friendly and approachable interface designed to create a comfortable space for emotional interaction.

C. Performance Testing

To validate the system, we measured the response latency, the accuracy of the sensor fusion, and the synchronization timing between the biometric sensors and the AI processing pipeline. By offloading the heavy neural network tasks to the *Hailo-8 NPU*, the system achieves a near-instantaneous response, which is vital for maintaining the flow of a natural conversation.

TABLE I
SYSTEM VALIDATION METRICS

Metric	Target	Measured
Response Latency	< 2.0s	1.4s
Stress Recognition	$> 85\%$	87%
Sensor Sync	< 1.0s	0.6s

V. CONCLUSION

The Nabu project successfully demonstrates the potential of hybrid AI architectures in the field of emotional support. By combining the local processing power of the *Hailo-8 accelerator* with the linguistic depth of a *therapy-trained LLM*, the system provides a responsive and secure environment for user interaction.

The transition from a technical tool to a supportive companion is validated by the research in [9], which emphasizes that for an AI to be effective, its interface must express "human-like" cues that foster user trust and comfort. In the case of Nabu, this is achieved through the *Nabu's Room* interface, which reduces the perceived distance between the user and the technology. Furthermore, the modular architecture of the system aligns with the principles of social robotics established in [10], which state that an intelligent agent must be both physically and emotionally accessible to the user.

Ultimately, Nabu stands as a functional prototype for a new generation of empathetic technology, designed to prioritize both technical excellence and the user's emotional well-being.

ACKNOWLEDGMENT

Gemini was utilized for technical writing refinement and LaTeX formatting. All scientific content, experimental results, and architectural logic are original and produced by the authors.

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